

ADO.NET and Entity Framework Core CRUD Programming Questions

Question 1: ADO.NET CRUD Operation using SQL Server

Write a C# console application using ADO.NET to perform CRUD operations on a Student database table.

Requirements:

Create a database table named Student2 with the following fields:

- Id (int, Primary Key)
- Name (varchar)
- Age (int)

Write a C# program that:

1. Establishes a connection with SQL Server database using SqlConnection.
2. Inserts a new student record into the Student2 table.
3. Retrieves and displays all student records using SqlDataReader.
4. Updates the age of a student record based on Id.
5. Deletes a student record based on Id.
6. Displays appropriate success messages after each operation.

Technologies:

- C#
- ADO.NET
- SQL Server
- SqlConnection
- SqlCommand
- SqlDataReader

Question 2: Entity Framework Core CRUD Operation

Create a C# console application using Entity Framework Core to perform CRUD operations on a Student entity.

Requirements:

Create a model class named Student with:

- Id
- Name

- Age

Create a database context class named CollegeContext that:

1. Inherits from DbContext.
2. Contains DbSet.
3. Configures SQL Server database connection.

Perform the following operations:

1. Insert Operation:

- Create a Student object.
- Add the object into the Student table.
- Save changes.

2. Read Operation:

- Retrieve all student records using LINQ.
- Display student details.

3. Update Operation:

- Find a student record using Id.
- Modify Age.
- Save changes.

4. Delete Operation:

- Find a student record.
- Remove it from database.
- Save changes.

Technologies:

- C#
- Entity Framework Core
- DbContext
- DbSet
- LINQ
- SQL Server

Combined Exam Question:

Develop a C# console-based Student Management System using both ADO.NET and Entity Framework Core.

The system should connect with SQL Server database named CollegeDB and maintain student information.

Implement:

1. Add a student.
2. Display all students.
3. Update student information.
4. Delete a student.

First implement CRUD operations using ADO.NET, then implement the same operations using Entity Framework Core.

Explain the difference between ADO.NET and Entity Framework Core based on:

- Database connectivity
- Query writing
- Data mapping
- Code complexity
- Performance